

3DTopia-XL: Scaling High-quality 3D Asset Generation via Primitive Diffusion

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<https://3dtopia.github.io/3DTopia-XL/>

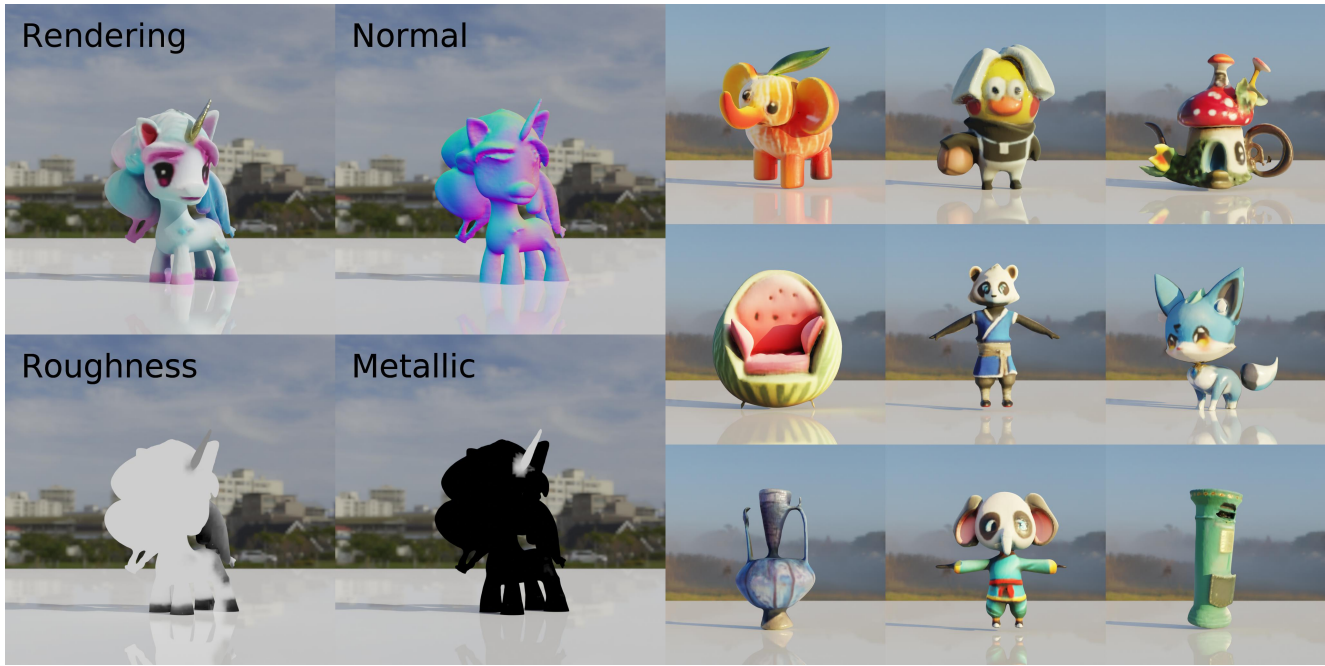


Figure 1. 3DTopia-XL generates high-quality 3D assets with smooth geometry and spatially varied textures and materials. The output asset (GLB mesh) can be seamlessly ported into graphics engines for physically-based rendering.

Abstract

The increasing demand for high-quality 3D assets across various industries necessitates efficient and automated 3D content creation. Despite recent advancements in 3D generative models, existing methods still face challenges with optimization speed, geometric fidelity, and the lack of assets for physically based rendering (PBR). In this paper, we introduce 3DTopia-XL, a scalable native 3D generative model designed to overcome these limitations. 3DTopia-XL leverages a novel primitive-based 3D representation, PrimX, which encodes detailed shape, albedo, and material field into a compact tensorial format, facilitating the modeling of high-resolution geometry with PBR assets. On top of the

novel representation, we propose a generative framework based on Diffusion Transformer (DiT), which comprises 1) Primitive Patch Compression, 2) and Latent Primitive Diffusion. 3DTopia-XL learns to generate high-quality 3D assets from textual or visual inputs. Extensive qualitative and quantitative evaluations are conducted to demonstrate that 3DTopia-XL significantly outperforms existing methods in generating high-quality 3D assets with fine-grained textures and materials, efficiently bridging the quality gap between generative models and real-world applications.

1. Introduction

High-quality 3D assets are essential for various real-world applications like films, gaming, and virtual reality. How-

ever, creating high-quality 3D assets involves extensive manual labor and expertise. Therefore, it further fuels the demand for automatic 3D content creation techniques, which automatically generate 3D assets from visual or textual inputs by using 3D generative models.

Fortunately, rapid progress has been witnessed in the field of 3D generative models recently. Existing state-of-the-art techniques can be sorted into three categories. **1)** Methods based on Score Distillation Sampling (SDS) [37, 44] lift 2D diffusion priors into 3D representation by per-scene optimization. However, these methods suffer from time-consuming optimization, poor geometry, and multifaceted inconsistency. **2)** Methods based on sparse-view reconstruction [14, 54] that leverage large models to regress 3D assets from single- or multi-view images. Most of these methods are built upon triplane-NeRF [3] representation. However, due to the triplane’s parameter inefficiency, the valid parameter space is limited to low resolutions in those models, leading to relatively low-quality 3D assets. Plus, reconstruction-based models also suffer from a low-diversity problem as deterministic methods. **3)** Methods as native 3D generative models [23, 58] aim to model the probabilistic distribution of 3D assets, generating 3D objects given input conditions. Yet, few of them are capable of generating high-quality 3D objects with Physically Based Rendering (PBR) assets, which are geometry, texture, and material packed into a GLB file.

To address the limitations above, we propose 3DTopia-XL, a high-quality native 3D generative model for 3D assets at scale. Our key idea is scaling the powerful diffusion transformer [36] on top of a novel primitive-based 3D representation. At the core of 3DTopia-XL is an efficient 3D representation, PrimX, which encodes the shape, albedo, and material of a textured mesh in a compact $N \times D$ tensor, enabling the modeling of high-resolution geometry with PBR assets. In specific, we anchor N primitives to the positions sampled on the mesh surface. Each primitive is a tiny voxel, parameterized by its 3D position, a global scale factor, and corresponding spatially varied payload for SDF, RGB, and material. Note that the proposed representation differentiates itself from the shape-only representation M-SDF [58] that PrimX encodes shape, color, and material in a unified way. It also supports efficient differentiable rendering, leading to the great potential to learn from not only 3D data but also image collections. Moreover, we carefully design initialization and fine-tuning strategy that enables PrimX to be rapidly tensorized from a textured mesh (GLB file) which is ten times faster than the triplane under the same setting.

Thanks to the tensorial and compact PrimX, we scale the 3D generative modeling using latent primitive diffusion with Transformers, where we treat each 3D object as a set of primitives. In specific, the proposed 3D generation framework consists of two modules. **1)** Primitive Patch Compres-

sion uses a 3D VAE for spatial compression of each individual primitive to get latent primitive tokens; and **2)** Latent Primitive Diffusion leverages the Diffusion Transformers (DiT) [36] to model global correlation of latent primitive tokens for generative modeling. The significant efficiency of the proposed representation allows us to achieve high-resolution generative training using a clean and unified framework without super-resolution to upscale the underlying 3D representation or post-hoc optimization-based mesh refinement.

In addition, we also carefully design algorithms for high-quality 3D PBR asset extraction from PrimX, to ensure reversible transformations between PrimX and textured mesh. An issue for most 3D generation models [49, 54] is that they use vertex coloring to represent the object’s texture, leading to a significant quality drop when exporting their generation results into mesh format. Thanks to the high-quality surface modeled by Signed Distance Field (SDF) in PrimX, we propose to extract the 3D shape with zero-level contouring and sample texture and material values in a high-resolution UV space. This leads to high-quality asset extraction with considerably fewer vertices, which is also ready to be packed into GLB format for downstream tasks.

Extensive experiments are conducted both qualitatively and quantitatively to evaluate the effectiveness of our method in text-to-3D and image-to-3D tasks. Moreover, we do extensive ablation studies to motivate our design choices for a better efficiency-quality tradeoff in the context of generative modeling with PrimX. In conclusion, we summarize our contributions as follows: **1)** We propose a novel 3D representation, PrimX, for high-quality 3D content creation, which is efficient, tensorial, and renderable. **2)** We introduce a scalable generative framework, 3DTopia-XL, tailored for generating high-quality 3D assets with high-resolution geometry, texture, and materials. **3)** Practical techniques for assets extraction from 3D representation to avoid quality gap. **4)** We demonstrate the superior quality and impressive applications of 3DTopia-XL for image-to-3D and text-to-3D tasks.

2. Related Work

Sparse-view Reconstruction Models. Recent advancements have been focusing on deterministic reconstruction methods that regress 3D assets from single- or multi-view images. Large Reconstruction Model (LRM) [10, 14] has shown that end-to-end training of a triplane-NeRF [3] regression model scales well to large datasets and can be highly generalizable. Although it can significantly accelerate generation speed, the generated 3D assets still exhibit relatively lower quality due to representation inefficiency and suffer from a low-diversity problem as a deterministic method. Subsequent works have extended this method to improve generation quality. For example, us-

ing multi-view images [1, 15, 20, 43, 47, 49, 52, 54] generated by 2D diffusion models as the input can effectively enhance the visual quality. However, the generative capability is actually enabled by the frontend multi-view diffusion models [22, 25, 29, 42] which cannot produce multi-view images with accurate 3D consistency. Another direction is to use more efficient 3D representations such as Gaussian Splatting [4, 17, 45, 56, 59, 62] and triangular mesh [21, 27, 50, 61, 66]. However, few of them can generate high-quality PBR assets with sampling diversity.

Native 3D Diffusion Models. Similar to 2D diffusion models for image generation, efforts have been made to train a 3D native diffusion model [7, 24, 39, 65] on conditional 3D generation. However, unlike the universal image representation in 2D, there are many different choices for 3D representations. Voxel-based methods [32] can be directly extended from 2D methods, but they are constrained by the demanding memory usage, and suffer from scaling up to high-resolution data. Point cloud based methods [33, 34] are memory-efficient and can adapt to large-scale datasets, but they hardly represent the watertight and solid surface of the 3D assets. Implicit representations such as triplane-NeRF offer a better balance between memory and quality [2, 5, 7, 9, 16, 26, 35, 48]. There are also methods based on other representations such as meshes and primitives [6, 28, 55, 57, 58]. However, these methods still struggle with generalization or producing high-quality assets. Recent methods attempt to adapt latent diffusion models to 3D [13, 19, 23, 46, 51, 60, 63, 64]. These methods first train a 3D compression model such as a VAE to encode 3D assets into a more compact form, which allows the diffusion model to train more effectively and show strong generalization. However, they either suffer from low-resolution results or are incapable of modeling PBR materials. In this paper, we propose a new 3D latent diffusion model on a novel representation, PrimX, which can be efficiently computed from a textured mesh and unpacked into high-resolution geometry with PBR materials.

3. Methodology

3.1. PrimX: An Efficient Representation for Shape, Texture, and Material

Before diving into details, we outline the following design principles for 3D representation in the context of high-quality large-scale 3D generative models: **1) Parameter-efficient:** provides a good trade-off between approximation error and parameter count; **2) Rapidly tensorizable:** can be efficiently transformed into a tensor, which facilitates generative modeling with modern neural architectures; **3) Differentially renderable:** compatible with differentiable renderer, enabling learning from both 3D and 2D data.

Given the aforementioned principles, we propose a novel

primitive-based 3D representation, namely PrimX, which represents the 3D shape, texture, and material of a textured mesh as a compact $N \times D$ tensor. It can be efficiently computed from a textured mesh (in GLB format) and directly rendered into 2D images via a differentiable rasterizer.

3.1.1. Definition

Preliminaries. Given a textured 3D mesh, we denote its 3D shape as $\mathcal{S} \in \mathbb{R}^3$, where $\mathbf{x} \in \mathcal{S}$ are spatial points inside the occupancy of the shape, and $\mathbf{x} \in \partial\mathcal{S}$ are the points on the shape’s boundary, *i.e.* the shape’s surface. We model the 3D shape as SDF as follows:

$$F_{\mathcal{S}}^{\text{SDF}}(\mathbf{x}) = \text{sign}(\mathbf{x}, \mathcal{S}) \cdot d(\mathbf{x}, \partial\mathcal{S}), \quad (1)$$

where $d(\mathbf{x}, \partial\mathcal{S}) = \min_{\mathbf{y} \in \mathcal{S}} \|\mathbf{x} - \mathbf{y}\|_2$ is the distance function, $\text{sign}(\mathbf{x}, \mathcal{S}) = 1$ when $\mathbf{x} \in \mathcal{S}$, and $\text{sign}(\mathbf{x}, \mathcal{S}) = -1$ when $\mathbf{x} \notin \mathcal{S}$. Moreover, given the neighborhood of the surface, $\mathcal{U}(\partial\mathcal{S}, \delta) = \{d(\mathbf{x}, \partial\mathcal{S}) < \delta\}$, the space-varied color and material functions of the target mesh are defined:

$$F_{\mathcal{S}}^{\text{RGB}}(\mathbf{x}) = \text{sign}(\mathbf{x}, \mathcal{U}) \cdot C(\mathbf{x}), F_{\mathcal{S}}^{\text{Mat}}(\mathbf{x}) = \text{sign}(\mathbf{x}, \mathcal{U}) \cdot \rho(\mathbf{x}), \quad (2)$$

where $\text{sign}(\mathbf{x}, \mathcal{U}) = 1$ when $\mathbf{x} \in \mathcal{U}$, and $\text{sign}(\mathbf{x}, \mathcal{U}) = 0$ when $\mathbf{x} \notin \mathcal{U}$. The $C(\mathbf{x}) : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ and $\rho(\mathbf{x}) : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ are corresponding texture sampling functions to get albedo and material (metallic and roughness) from UV-aligned texture maps given the 3D point $\mathbf{x}$. Eventually, all shape, texture, and material information of a 3D mesh can be parameterized by the volumetric function $F_{\mathcal{S}} = (F_{\mathcal{S}}^{\text{SDF}} \oplus F_{\mathcal{S}}^{\text{RGB}} \oplus F_{\mathcal{S}}^{\text{Mat}}) : \mathbb{R}^3 \rightarrow \mathbb{R}^6$, where $\oplus$ denotes concatenation.

PrimX Representation. We aim to approximate $F_{\mathcal{S}}$ with a neural volumetric function $F_{\mathcal{V}} : \mathbb{R}^3 \rightarrow \mathbb{R}^6$ parameterized by a $N \times D$ tensor $\mathcal{V}$. For efficiency, our key insight is to define $F_{\mathcal{V}}$ as a set of N volumetric primitives distributed on the surface of the mesh:

$$\mathcal{V} = \{\mathcal{V}_k\}_{k=1}^N, \text{ where } \mathcal{V}_k = \{\mathbf{t}_k, s_k, \mathbf{X}_k\}. \quad (3)$$

Each primitive $\mathcal{V}_k$ is a tiny voxel with a resolution of a^3 , parameterized by its 3D position $\mathbf{t}_k \in \mathbb{R}^3$, a global scale factor $s_k \in \mathbb{R}^+$, and corresponding spatially varied feature payload $\mathbf{X}_k \in \mathbb{R}^{a \times a \times a \times 6}$ within the voxel. Note that, the payload in PrimX could be spatially varied features with any dimensions. Our instantiation here is to use a six-channel local grid $\mathbf{X}_k = \{\mathbf{X}_k^{\text{SDF}}, \mathbf{X}_k^{\text{RGB}}, \mathbf{X}_k^{\text{Mat}}\}$ to parameterize SDF, RGB color, and material respectively.

Inspired by Yariv et al. [58] where mosaic voxels are globally weighted to get a smooth surface, the approximation of a textured mesh is then defined as a weighted combination of primitives:

$$F_{\mathcal{V}}(\mathbf{x}) = \sum_{k=1}^N [w_k(\mathbf{x}) \cdot \mathcal{I}(\mathbf{X}_k, (\mathbf{x} - \mathbf{t}_k)/s_k)], \quad (4)$$

where $\mathcal{I}(\mathcal{V}_k, \mathbf{x})$ denotes the trilinear interpolant over the voxel grid $\mathbf{X}_k$ at position $\mathbf{x}$. The weighting function $w_k(\mathbf{x})$

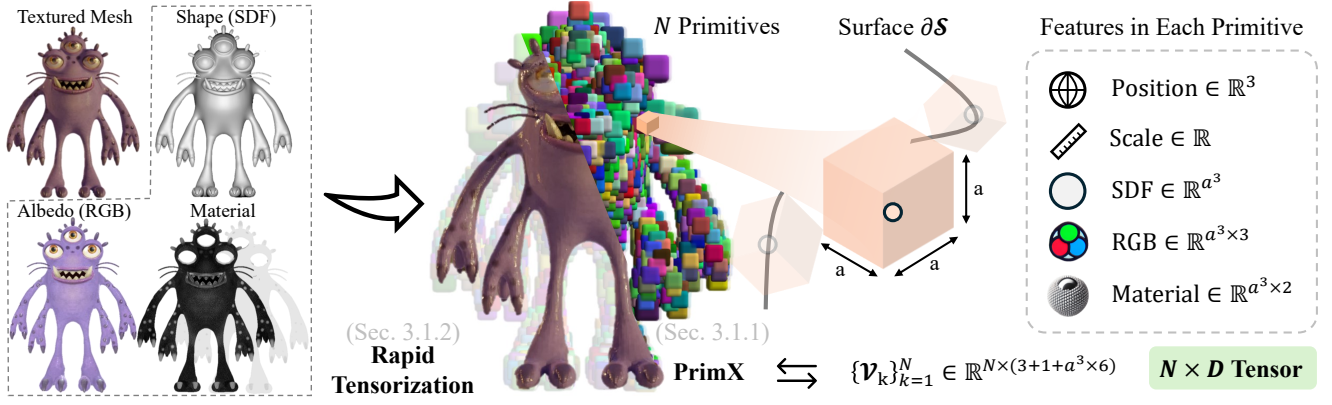


Figure 2. **Illustration of PrimX.** We propose to represent the 3D shape, texture, and material of a textured mesh as a compact $N \times D$ tensor (Sec. 3.1.1). We anchor N primitives to the positions sampled on the mesh surface. Each primitive $\mathcal{V}_k$ is a tiny voxel with a resolution of a^3 , parameterized by its 3D position $\mathbf{t}_k \in \mathbb{R}^3$, a global scale factor $s_k \in \mathbb{R}^+$, and corresponding spatially varied payload $\mathbf{X}_k \in \mathbb{R}^{a \times a \times a \times 6}$ for SDF, RGB, and material. This tensorial representation can be rapidly computed from a textured mesh within 1.5 minutes (Sec. 3.1.2).

of each primitive is defined as:

$$w_k(\mathbf{x}) = \frac{\hat{w}_k(\mathbf{x})}{\sum_{j=1}^N \hat{w}_j(\mathbf{x})}, \quad (5)$$

$$\text{s.t. } \hat{w}_k(\mathbf{x}) = \max(0, 1 - \|\frac{\mathbf{x} - \mathbf{t}_k}{s_k}\|_\infty). \quad (6)$$

Once the payload of primitives is determined, we can leverage a highly efficient differentiable renderer to turn PrimX into 2D images. In specific, given a camera ray $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$ with camera origin $\mathbf{o}$ and ray direction $\mathbf{d}$, the corresponding pixel value I is solved by the following integral:

$$I = \int_{t_{\min}}^{t_{\max}} F_{\mathcal{V}}^{\text{RGB}}(\mathbf{r}(t)) \frac{dT(t)}{dt} dt, \quad (7)$$

where $T(t)$ is an exponential function of the SDF field to represent the opacity field:

$$T(t) = \int_{t_{\min}}^t \exp[-(\frac{F_{\mathcal{V}}^{\text{SDF}}(\mathbf{r}(t))}{\alpha})^2] dt. \quad (8)$$

And $\alpha = 0.005$ is the hyperparameter that controls the variance of the opacity field during this conversion.

To wrap up, the learnable parameters of a textured 3D mesh modeled by PrimX are primitive position $\mathbf{t} \in \mathbb{R}^{N \times 3}$, primitive scale $s \in \mathbb{R}^{N \times 1}$, and voxel payload $\mathbf{X} \in \mathbb{R}^{N \times a^3 \times 6}$ for SDF, albedo, and material. Therefore, each textured mesh can be represented as a compact $N \times D$ tensor, where $D = 3 + 1 + a^3 \times 6$ by concatenation.

PBR Asset Extraction. Once PrimX is constructed, it encodes all geometry and appearance information of the target mesh within the $N \times D$ tensor. Now, we introduce our efficient algorithm to convert PrimX back into a textured mesh in GLB file format. For geometry, we can easily extract the corresponding 3D shape with Marching Cubes algorithm [30] on zero level set of $F_{\mathcal{V}}^{\text{SDF}}$. For PBR texture maps, we first perform UV unwrapping in a high-resolution

UV space (1024×1024). Then, we get sampling points in 3D and query $\{F_{\mathcal{V}}^{\text{RGB}}, F_{\mathcal{V}}^{\text{Mat}}\}$ to get corresponding albedo and material values. Note that, we mask the UV space to get the index of valid vertices for efficient queries. Moreover, we dilate the UV texture maps and inpaint the dilated region with the nearest neighbors of existing textures, ensuring albedo and material maps smoothly blend outwards for anti-aliasing. Finally, we pack geometry, UV mapping, albedo, and material maps into a GLB file, which is ready for the graphics engine and various downstream tasks.

3.1.2. Computing PrimX from Textured Mesh

We introduce our efficient fitting algorithm in this section that computes PrimX from the input textured mesh in a short period of time so that it is scalable on large-scale datasets for generative modeling. Given a textured 3D mesh F_S , our goal is to compute PrimX such that $F_{\mathcal{V}}(\mathbf{x}) \approx F_S(\mathbf{x})$, s.t. $\mathbf{x} \in \mathcal{U}(\partial S, \delta)$. Our key insight is that the fitting process can be efficiently achieved via a good initialization followed by lightweight finetuning.

Initialization. We assume all textured meshes are provided in GLB format which contains triangular meshes, texture and material maps, and corresponding UV mappings. The vertices of the target mesh are first normalized within the unit cube. To initialize the position of primitives, we first apply uniform random sampling on the mesh surface to get N candidate initial points. Then, we perform farthest point sampling on this candidate point set to get N valid initial positions for all primitives. This two-step initialization of position ensures good coverage of $F_{\mathcal{V}}$ over the boundary neighborhood $\mathcal{U}$ while also keeping the high-frequency shape details as much as possible. Then, we compute the L2 distance of each primitive to its nearest neighbors, taking the value as the initial scale factor for each primitive.

To initialize the payload of primitives, we first compute candidate points in global coordinates using initialized positions $\mathbf{t}_k$ and scales s_k as $\mathbf{t}_k + s_k \mathbf{I}$ for each primitive,

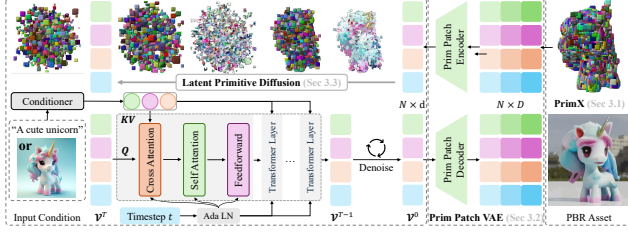


Figure 3. **Overview of 3DTopia-XL.** As a native 3D diffusion model, 3DTopia-XL is built upon a novel 3D representation PrimX (Sec. 3.1). This compact and expressive representation encodes the shape, texture, and material of a textured mesh efficiently, which allows modeling high-resolution geometry with PBR assets. Furthermore, this tensorial representation facilitates our patch-based compression using primitive patch VAE (Sec. 3.2). We then use our novel latent primitive diffusion (Sec. 3.3) for 3D generative modeling, which operates the diffusion and denoising process on the set of latent PrimX, naturally compatible with Transformer-based neural architectures.

where $\mathbf{I}$ is the unit local voxel grid with a resolution of a^3 . To initialize the SDF value, we query the SDF function converted from the 3D shape at each candidate point, *i.e.* $\mathbf{X}_k^{\text{SDF}} = F_S^{\text{SDF}}(\mathbf{t}_k + s_k \mathbf{I})$. Notably, it is non-trivial to get a robust conversion from arbitrary 3D shape to volumetric SDF function. Our implementation is based on an efficient ray marching with bounding volume hierarchy that works well with non-watertight topology. To initialize the color and material values, we sample the corresponding albedo colors and material values from UV space using geometric functions F_S^{RGB} and F_S^{Mat} . Specifically, we compute the closest face and corresponding barycentric coordinates for each candidate point on the mesh, then interpolate UV coordinates and sample from the texture maps to get the value. **Finetuning.** Even if the initialization above offers a fairly good estimate of F_S , a rapid finetuning process can further decrease the approximation error via gradient descent. Specifically, we optimize the well-initialized PrimX with a regression-based loss on SDF, albedo, and material values:

$$\begin{aligned} \mathcal{L}(\mathbf{x}; \mathcal{V}) = & \lambda_{\text{SDF}} \|F_S^{\text{SDF}}(\mathbf{x}) - F_{\mathcal{V}}^{\text{SDF}}(\mathbf{x})\|_1 \\ & + \lambda (\|F_S^{\text{RGB}}(\mathbf{x}) - F_{\mathcal{V}}^{\text{RGB}}(\mathbf{x})\|_1 \\ & + \|F_S^{\text{Mat}}(\mathbf{x}) - F_{\mathcal{V}}^{\text{Mat}}(\mathbf{x})\|_1), \end{aligned} \quad (9)$$

where $\forall \mathbf{x} \in \mathcal{U}$, and $\lambda_{\text{SDF}}, \lambda$ are loss weights. We employ a two-stage finetuning strategy where we optimize with $\lambda_{\text{SDF}} = 10$ and $\lambda = 0$ for the first 1k iterations and $\lambda_{\text{SDF}} = 0$ and $\lambda = 1$ for the second 1k iterations. More details are provided in the supplementary document.

3.2. Primitive Patch Compression

In this section, we introduce our lightweight patch-based compression on individual primitives that turns 3D primitives into latent tokens for efficient generative modeling.

We opt for using a variational autoencoder [18] (VAE) operating on local voxel patches which compresses the payload of each primitive into latent tokens, *i.e.* $F_{\text{ae}} : \mathbb{R}^D \rightarrow \mathbb{R}^d$. Specifically, the autoencoder F_{ae} consists of an encoder E and a decoder D building with 3D convolutional layers. The encoder F_{ae} has a downsampling rate of 48 that compresses the voxel payload $\mathbf{X}_k \in \mathbb{R}^{a^3 \times 6}$ into the voxel latent $\hat{\mathbf{X}}_k \in \mathbb{R}^{(a/2)^3 \times 1}$. We train F_{ae} with reconstruction loss:

$$\mathcal{L}_{\text{ae}}(\mathbf{X}; E, D) = \mathbb{E}[\|\mathbf{X}_k - D(E(\mathbf{X}_k))\|_2 + \lambda_{\text{kl}} \mathcal{L}_{\text{kl}}(\mathbf{X}_k, E)], \quad (10)$$

where λ_{kl} is the weight for KL regularization over the latent space. Note that, unlike other works on 2D/3D latent diffusion models [40, 63] that perform global compression over all patches, our VAE only compresses each local primitive patch independently and defers the modeling of global semantics and inter-patch correlation to the diffusion model. Once the VAE is trained, we can compress the raw PrimX as $\mathcal{V}_k = \{\mathbf{t}_k, s_k, E(\mathbf{X}_k)\}$. It leads to a low-dimensional parameter space for the diffusion model as $\mathcal{V} \in \mathbb{R}^{N \times d}$, where $d = 3 + 1 + (a/2)^3$. In practice, this compact parameter space significantly allows more model parameters given a fixed computational budget, which is the key to scaling up 3D generative models in high resolution.

3.3. Latent Primitive Diffusion

On top of PrimX (Sec. 3.1) and the corresponding VAE (Sec. 3.2), the problem of 3D object generation is then converted to learning the distribution $p(\mathcal{V})$ over large-scale datasets. Our goal is to train a diffusion model [12] that takes as input random noise $\mathcal{V}^T$ and conditions $\mathbf{c}$, and predicts PrimX samples. Note that, the target space for denoising is $\mathcal{V}^T \in \mathbb{R}^{N \times d}$, where $d = 3 + 1 + (a/2)^3$.

In specific, the diffusion model learns to denoise $\mathcal{V}^T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ through denoising steps $\{\mathcal{V}^{T-1}, \dots, \mathcal{V}^0\}$ given conditional signal $\mathbf{c}$. As a set of primitives, PrimX is naturally compatible with Transformer-based architectures, where we treat each primitive as a token. Moreover, the permutation equivariance of PrimX removes the need for any positional encoding in Transformers.

Our largest latent primitive diffusion model g_{Φ} is a 28-layer transformer, with cross-attention layers to incorporate conditional signals, self-attention layers for modeling inter-primitive correlations, and adaptive layer normalization to inject timestep conditions. The model g_{Φ} learns to predict at timestep t given input condition signal:

$$g_{\Phi}(\mathcal{V}^t, t, \mathbf{c}) = \{\text{AdaLN}[\text{SelfAttn}(\text{CrossAttn}(\mathcal{V}^t, \mathbf{c}, \mathbf{c})), t]\}^{28}, \quad (11)$$

where $\text{CrossAttn}(\mathbf{q}, \mathbf{k}, \mathbf{v})$ denotes the cross-attention layer with query, key, and value as input. $\text{SelfAttn}(\cdot)$ denotes the self-attention layer. $\text{AdaLN}(\cdot, t)$ denotes adaptive layer normalization layers to inject timestep conditioned

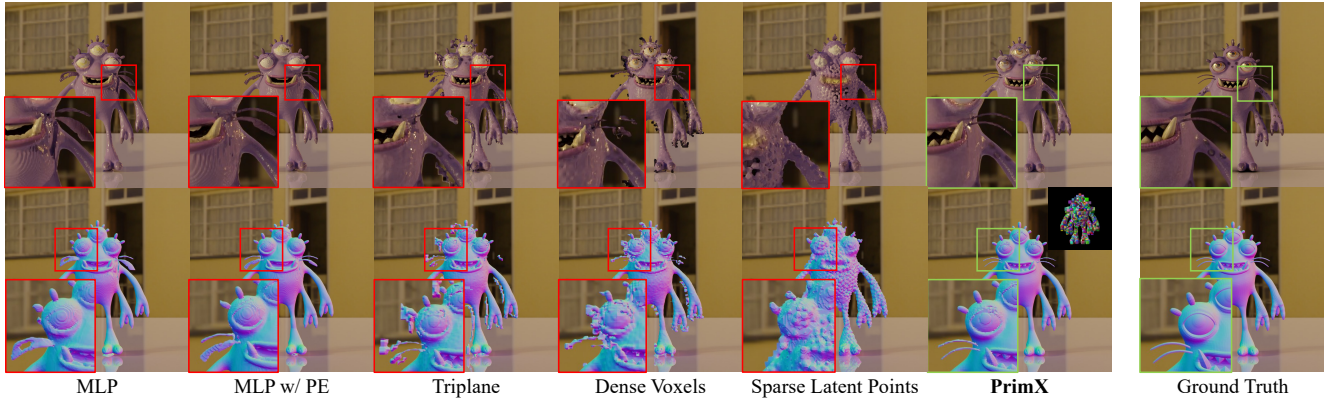


Figure 4. **Evaluations of different 3D representations.** We evaluate the effectiveness of different representations in fitting the ground truth’s shape, texture, and material (right). All representations are constrained to a budget of 1.05M parameters. PrimX achieves the highest fidelity in terms of geometry and appearance with significant strength in runtime efficiency (Table 1) at the same time.

Table 1. **Quantitative evaluations of different 3D representations.** We evaluate the approximation error of different representations for shape, texture, and material. All representations adhere to a parameter budget of 1.05M. PrimX shows the best fitting quality, especially for the geometry (also shown in Figure 4), while having the most speedy fitting runtime. The top three techniques are highlighted in red, orange, and yellow, respectively.

Representation	Runtime	$CD \times 10^{-4} \downarrow$	$PSNR-F_S^{SDF} \uparrow$	$PSNR-F_S^{RGB} \uparrow$	$PSNR-F_S^{Mat} \uparrow$
MLP	14 min	4.502	40.73	21.19	13.99
MLP w/ PE	14 min	4.638	40.82	21.78	12.75
Triplane	16 min	9.678	39.88	18.28	16.46
Dense Voxels	10 min	7.012	41.70	20.01	15.98
PrimX	1.5 min	1.310	41.74	21.86	16.50

modulation to cross-attention, self-attention, and feed-forward layers. Moreover, we employ the pre-normalization scheme [53] for training stability. For noise scheduling, we use discrete 1,000 noise steps with a cosine scheduler during training. We opt for “v-prediction” [41] with Classifier-Free Guidance (CFG) [11] as the training objective for better conditional generation quality and faster convergence:

$$\mathcal{L}_{\text{diff}}(\Phi) = \mathbb{E}_{t \sim [1, T], \mathcal{V}^0, \mathcal{V}^t} [\| (\sqrt{\bar{\alpha}_t} \epsilon - \sqrt{1 - \bar{\alpha}_t} \mathcal{V}^0) - g_{\Phi}(\mathcal{V}^t, t, \bar{\mathbf{c}}(b)) \|_2^2], \quad (12)$$

where ϵ is the noise sampled from Gaussian distribution, $\bar{\alpha}_t = \prod_{i=0}^t (1 - \beta_i)$ and β_t comes from our cosine beta scheduler. And $b \sim \mathcal{B}(p_0)$ is a random variable sampled from Bernoulli distribution taking 0, 1 with probability p_0 and $1 - p_0$ respectively. And the condition signal under CFG is defined as $\bar{\mathbf{c}}(b) = b \cdot \mathbf{c} + (1 - b) \cdot \emptyset$, where $\emptyset$ is the learnable embedding for unconditional generation.

4. Experiments

4.1. Representation Evaluation

Evaluation Protocol. We first evaluate different designs of 3D representations in the context of 3D generative modeling. Our evaluation principles focus on two aspects: 1) runtime from GLB mesh to the representation, and 2) approximation error for shape, texture, and material given a fixed computational budget. Given 30 GLB meshes ran-

domly sampled from our training dataset, we take the average fitting time till convergence as runtime measured on an A100 GPU. For geometry quality, we evaluate the Chamfer Distance (CD) and the Peak Signal-to-Noise Ratio (PSNR) of SDF values between ground truth meshes and fittings. For appearance quality, we evaluate the PSNR of albedo and materials values. All metrics are computed on a set of 500k points sampled near the shape surface.

Baselines. Given our final hyperparameters of PrimX, where $N = 2048, a = 8$, we fix the number of parameters of all representations to $2048 \times 8^3 \approx 1.05\text{M}$ for comparisons. We compare four alternative representations: 1) **MLP**: a pure Multi-Layer Perceptron with 3 layers and 1024 hidden dimensions; 2) **MLP w/ PE**: the MLP baseline with Positional Encoding (PE) [31] to the input coordinates; 3) **Triplane** [3]: three orthogonal 2D planes with a resolution of 128×128 and 16 channels, followed by a two-layer MLP decoder with 512 hidden dimensions. 4) **Dense Voxels**: a dense 3D voxel with a resolution of $100 \times 100 \times 100$. 5) **Sparse Points**: 2048 latent points followed by a three-layer MLP as the decoder. All methods are trained with the same objective (Eq. 9) and points sampling strategy as ours.

Results. Quantitative results are presented in Table 1, which shows that PrimX achieves the least approximation error among all methods, especially for geometry (indicated by CD). Besides the best quality, the proposed representation demonstrates significant efficiency in terms of runtime with nearly 7 times faster convergence speed compared with the second best, making it scalable on large-scale datasets. Figure 4 shows qualitative comparisons. MLP-based methods have periodic artifacts, especially for the geometry. Triplane and dense voxels yield bumpy surfaces and grid artifacts. Instead, PrimX produces the best quality with smooth geometry and fine-grained details like the tapering beard.

4.2. Image-to-3D Generation

Comparison Methods. We run evaluations against two types of methods: 1) **sparse-view reconstruction models**, and 2) **image-conditioned diffusion models**.



Figure 5. **Image-to-3D comparisons.** For each method, we take the textured mesh predicted from the input image into Blender and render it with the target environment map. We compare our single-view conditioned model with **sparse-view reconstruction models** and **image-conditioned diffusion models**. 3DTopia-XL achieves the best visual and geometry quality. Thanks to our capability to generate spatially varied PBR assets shown on the rightmost, our generated mesh can also produce vivid reflectance with specular highlights and glossiness.

The reconstruction-based models, like LGM [45], InstantMesh [54], Real3D [15], CRM [49], are deterministic methods that learn to reconstruct 3D objects given four or six input views. They enable single-view to 3D synthesis by leveraging pretrained diffusion models [22, 42] to generate multiple views from the input single image. However, those methods may suffer from multi-view inconsistency caused by the frontend 2D diffusion models. The feed-forward diffusion models, like CraftsMan [23], ShapE [16], LN3Diff [19], are probabilistic methods that learn to generate 3D objects given input image conditions. All methods above only model the shape and color without considering roughness and metallic while our method is suitable to produce those assets with sampling diversity. To fairly compare the quality in real-world use cases, we place the output mesh of each method into Blender [8] and render it with an environment map. For methods that cannot produce PBR materials, we assign the default diffuse material.

Results. Figure 5 demonstrates qualitative results. Existing reconstruction-based models fail to produce good results which may suffer from multiview inconsistency and incapability to support spatially varied materials. Moreover, most of reconstruction models are built upon triplane rep-

Table 2. **Analysis of number (N) and resolution (a) of PrimX.**

# Primitives	Resolution	# Parameters	PSNR- $F_S^{SDF} \uparrow$	PSNR- $F_S^{RGB} \uparrow$	PSNR- $F_S^{Mat} \uparrow$
$N = 64$	$a^3 = 32^3$	2.10M	61.05	22.18	18.10
$N = 256$	$a^3 = 16^3$	1.05M	59.05	23.50	18.61
$N = 512$	$a^3 = 8^3$	0.26M	59.57	22.58	18.50
$N = 512$	$a^3 = 16^3$	2.10M	62.89	23.92	18.21
$N = 2048$	$a^3 = 8^3$	1.05M	62.52	24.23	18.53

resentation which is not parameter-efficient. This downside limits the spatial resolution of the underlying 3D representation, leading to the bumpy surface indicated by the rendered normal. On the other hand, existing 3D diffusion models fail to generate objects that are visually aligned with the input condition. While CraftsMan is the only method with a comparable surface quality as ours, they are only capable of generating 3D shapes without textures and materials. In contrast, 3DTopia-XL achieves the best visual and geometry quality among all methods. Thanks to our capability to generate spatially varied PBR assets, our generated mesh can produce vivid reflectance with specular highlights even under harsh environmental illuminations.

4.3. Text-to-3D Generation

We conduct quantitative evaluations against native text-to-3D generative models, which achieves text-to-3D generation without text-to-multiview diffusion models. Given a

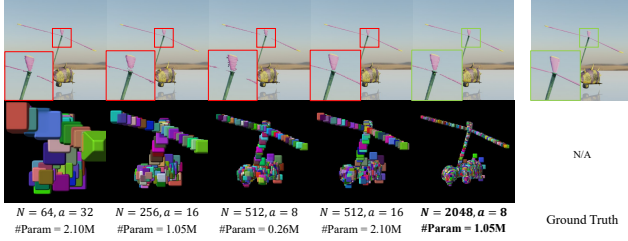


Figure 6. **Ablation studies of the number and resolution of primitives.** Our final setting ($N = 2048, a = 8$) has the optimal approximation quality of ground truth, especially for fine-grained details like thin rotor blades.

set of unseen text prompts, we take the CLIP Score [38] as the evaluation metric following previous work [13, 16]. We mainly compare two methods with open-source implementations: Shap-E [16] and 3DTopia [13]. As shown in Table 4, our method achieves better alignment between input text and rendering of the generated asset. We defer the qualitative results in the supplementary.

4.4. Further Analysis

Number and Resolution of Primitives. As a structured and serialized 3D representation, the number of primitives and the resolution of each primitive are two critical factors for the efficiency-quality tradeoff in PrimX. More and larger primitives often lead to better approximation quality. However, it results in a longer set length and deeper feature dimensions, causing inefficient long-context attention computation and training difficulty of the diffusion model. Therefore, we explore the impact of the number and resolution of primitives on different parameter budgets. We evaluate the PSNR of SDF, albedo, and material values given 500k points sampled near the surface. As shown in Table 2 and Figure 6, given a fixed parameter count, a larger set with compact primitives lead to better approximations. Furthermore, a longer sequence will increase the GFlops of DiT which also leads to better generation quality shown in the supplementary. Therefore, we use a large set of primitives with a relatively small local resolution.

Patch Compression Rate. The spatial compression rate of our primitive patch-based VAE (Sec. 3.2) is also an important design choice. Empirically, a higher compression rate leads to a more efficient latent diffusion model but risks information loss. Therefore, we analyze different compression rates given two different set lengths $N = 256$ and $N = 2048$ with the same parameter count of PrimX. We measure the PSNR between the VAE’s output and input on 1k random samples. Table 3 shows the results where the final choice of $N = 2048$ with compression rate $f = 48$ achieves the optimal VAE reconstruction. The setting with $N = 256, f = 48$ has the same compression rate but lower reconstruction quality and a latent space with higher resolution, which we find difficulty in the convergence of the

Table 3. **Analysis of different compression rates for VAE.** The scalar f stands for the compression rate between the input to the VAE and the latent code (*i.e.* the ratio between the 2nd and 3rd column of the table).

# Primitives	VAE input	Latent	f	PSNR $\uparrow$
$N = 256$	6×16^3	6×4^3	64	22.92
$N = 256$	6×16^3	1×4^3	384	19.80
$N = 256$	6×16^3	1×8^3	48	23.33
$N = 2048$	6×8^3	1×2^3	384	18.48
$N = 2048$	6×8^3	1×4^3	48	24.51

Table 4. **Text-to-3D Evaluations.** We evaluate the CLIP Score between input prompts and front-view renderings of output 3D assets.

Methods	CLIP Score $\uparrow$
ShapE	21.98
3DTopia	22.54
Ours	24.33

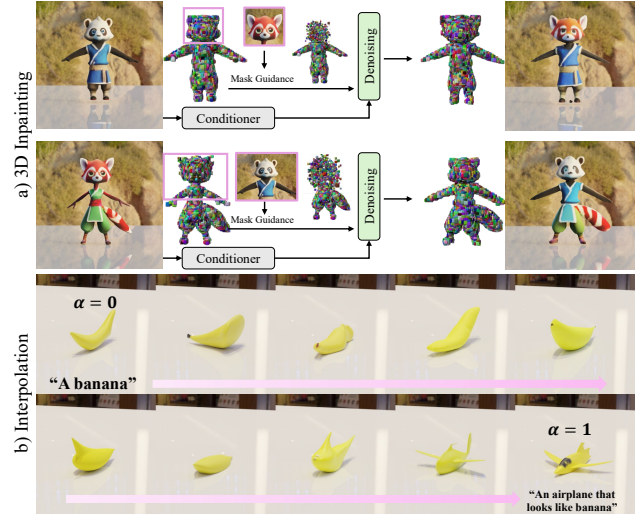


Figure 7. **3D Generative Applications.** a) 3D Inpainting for image-to-3D and b) Interpolation between text-to-3D samples.

latent primitive diffusion model g_{Φ} .

Inpainting and Interpolation. We show 3D generative applications like 3D inpainting and interpolation in Figure 7, showing the unique property of 3D native diffusion model compared to reconstruction methods.

Besides the ablation studies above, we also analyze 1) **the model scaling**, 2) **the sampling diversity**, 3) **PrimX initialization**, 4) **user study**, 5) **more comparisons** and 6) **more visual results**, which are deferred to the supplementary due to the space limit.

5. Conclusion

We present 3DTopia-XL, a native 3D diffusion model for PBR asset generation given textual or visual inputs. Central to our approach is PrimX, an innovative primitive-based 3D representation that is parameter-efficient, tensorial, and renderable. It encodes shape, albedo, and material into a compact $N \times D$ tensor, enabling the modeling of high-resolution geometry with PBR assets. On top of PrimX, we propose Latent Primitive Diffusion for scalable 3D generative models, together with practical techniques to export PBR assets ready for graphics pipelines.

Acknowledgement. This study is supported by the National Key R&D Program of China No.2022ZD0160102. This research/project is supported by the National Research Foundation, Singapore under its AI Singapore Programme (AISG Award No: AISG2-PhD-2021-08-019), the Ministry of Education, Singapore, under its MOE AcRF Tier 2 (MOET2EP20221- 0012, MOE-T2EP20223-0002), NTU NAP, and under the RIE2020 Industry Alignment Fund – Industry Collaboration Projects (IAF-ICP) Funding Initiative, as well as cash and in-kind contribution from the industry partner(s).

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